

Artificial Intelligence - Skill IQ Practice Question Bank

Medium, Hard, and Very Hard MCQs with answer key and explanations

Purpose: Build Skill IQ-style readiness using original scenario-based MCQs. These are not official Pluralsight exam questions or dumps.

Difficulty mix: Medium questions test core knowledge; Hard questions test applied decision-making; Very Hard questions test architecture, edge cases, trade-offs, and production judgement.

Focus areas

- AI definitions, symbolic AI, statistical AI, ML, deep learning, and generative AI
- Search, planning, agents, knowledge representation, NLP, computer vision, and reinforcement learning
- Training data, model evaluation, uncertainty, responsible AI, and governance
- AI implementation, operations, business use cases, and real-world risk management

How to use this bank

Do the questions timed first, then review the answer key. For Skill IQ-style prep, focus on why the correct answer is better than plausible alternatives.

Source scope

Aligned with public Pluralsight path descriptions and general industry knowledge. Topics emphasize breadth, applied reasoning, and production trade-offs rather than memorized question dumps.

Questions

Q01 - Medium - Definitions

Which statement best describes artificial intelligence?

- A. Only neural networks trained on images
- B. Systems that perform tasks associated with human intelligence such as reasoning, perception, learning, or language
- C. A database technology only
- D. A programming language

Q02 - Medium - AI vs ML

How is machine learning related to AI?

- A. ML is a subset of AI focused on learning patterns from data
- B. AI is a subset of ML
- C. They are unrelated
- D. ML only means robotics

Q03 - Medium - Deep learning

What distinguishes deep learning from many traditional ML methods?

- A. Use of multi-layer neural networks to learn representations
- B. No need for data
- C. Only works with symbolic rules
- D. Cannot be used for images

Q04 - Medium - Symbolic AI

Symbolic AI primarily relies on what?

- A. Explicit rules, logic, and knowledge representations
- B. Gradient descent only
- C. Random token sampling
- D. Image filters only

Q05 - Medium - Generative AI

What makes generative AI different from many predictive AI systems?

- A. It creates new content such as text, images, code, or audio
- B. It only sorts rows in a table
- C. It cannot use neural networks
- D. It never requires data

Q06 - Medium - Search

In AI search, what is a heuristic?

- A. A rule of thumb estimating distance or cost to a goal
- B. A label leakage technique
- C. A database password
- D. A model deployment server

Q07 - Medium - Agents

An AI agent is best described as a system that does what?

- A. Perceives an environment and takes actions toward goals
- B. Only stores files
- C. Only compresses images
- D. Cannot act

Q08 - Medium - NLP

Which is an NLP task?

- A. Sentiment analysis of reviews
- B. Object detection in images
- C. Motor control in a robot arm only
- D. Disk defragmentation

Q09 - Medium - Computer vision

Which task belongs mainly to computer vision?

- A. Detecting pedestrians in camera frames
- B. Translating English to French
- C. Parsing SQL queries
- D. Calculating interest rates

Q10 - Medium - Reinforcement learning

In reinforcement learning, what is a reward?

- A. Feedback signal that indicates how good an action or outcome is
- B. A training label for every pixel
- C. A SQL table
- D. An encryption key

Q11 - Medium - Data quality

Why is training data quality critical in AI?

- A. Models learn patterns from data, including errors and biases
- B. Data quality only affects storage cost
- C. Bad data always improves generalization
- D. AI systems do not use data

Q12 - Medium - Responsible AI

Which is a responsible AI concern?

- A. Fairness, transparency, privacy, accountability, and safety
- B. Only GPU speed
- C. Only font size
- D. Only UI color

Q13 - Hard - Knowledge representation

Why might a knowledge graph be useful in an AI system?

- A. It represents entities and relationships explicitly for reasoning and retrieval
- B. It replaces all data validation
- C. It makes neural networks unnecessary in every case
- D. It guarantees unbiased outputs

Q14 - Hard - Planning

What is the difference between planning and simple classification?

- A. Planning selects a sequence of actions to reach a goal; classification assigns labels
- B. Planning only uses images
- C. Classification requires rewards only
- D. They are identical

Q15 - Hard - Uncertainty

Why is uncertainty estimation important in AI decision systems?

- A. It helps decide when to abstain, escalate, or request more information
- B. It eliminates all wrong answers
- C. It increases model size automatically
- D. It replaces data governance

Q16 - Hard - AI lifecycle

Which sequence best reflects an AI project lifecycle?

- A. Problem framing -> data -> modeling -> evaluation -> deployment -> monitoring
- B. Deployment -> problem framing -> delete data
- C. Testing only -> production forever
- D. Collect all data -> no evaluation

Q17 - Hard - Evaluation

Why can high offline accuracy fail to translate into business value?

- A. The metric may not align with the real decision, cost, or workflow
- B. Accuracy is never useful
- C. Business value does not depend on AI systems
- D. The model has too few files

Q18 - Hard - Human-in-the-loop

When is human-in-the-loop review especially important?

- A. High-stakes, uncertain, regulated, or irreversible decisions
- B. Every harmless autocomplete suggestion
- C. Only when there is no data
- D. Never in AI systems

Q19 - Hard - Automation risk

What is automation bias?

- A. Humans over-trust automated recommendations even when they may be wrong
- B. Models becoming too small
- C. A faster GPU kernel
- D. A lack of internet access

Q20 - Hard - Generative AI risk

Why is hallucination a major risk in generative AI?

- A. Generated content can be plausible but unsupported or false
- B. It makes models unable to write
- C. It occurs only in images
- D. It means the model is encrypted

Q21 - Hard - AI operations

What is a key difference between deploying regular software and AI models?

- A. AI behavior can degrade as data distributions change
- B. AI code never has bugs
- C. AI systems need no monitoring
- D. Software cannot be tested

Q22 - Hard - Ethics

Why is explainability sometimes required in AI systems?

- A. Users, auditors, or regulators may need to understand reasons behind decisions
- B. It always improves accuracy
- C. It replaces cybersecurity
- D. It eliminates data collection

Q23 - Hard - Data privacy

Which practice best supports privacy in AI development?

- A. Data minimization, access control, anonymization/pseudonymization where suitable, and retention limits
- B. Copy all production data into prompts
- C. Store secrets in training data
- D. Share raw data with all teams

Q24 - Hard - Foundation models

Why are foundation models called foundation models?

- A. They are large general-purpose models that can be adapted to many downstream tasks
- B. They are always rule-based expert systems
- C. They can only classify images
- D. They never require compute

Q25 - Very Hard - Hybrid AI

Why might a hybrid symbolic-neural system be preferable to a purely neural system?

- A. It can combine learned perception/language with explicit rules, constraints, or reasoning
- B. It always has zero maintenance
- C. It removes the need for validation
- D. It guarantees no bias

Q26 - Very Hard - AI governance

An AI model is used to rank job applicants. What governance controls are most important?

- A. Bias testing, explainability, human review, documentation, privacy controls, and auditability
- B. Only increasing model size
- C. Only reducing latency
- D. No monitoring because hiring is subjective

Q27 - Very Hard - Distribution shift

An image classifier trained in daylight fails at night. What is the best diagnosis?

- A. Distribution shift between training and deployment conditions
- B. Perfect generalization
- C. Symbolic reasoning failure only
- D. Too much explainability

Q28 - Very Hard - RL tradeoffs

Why is reinforcement learning difficult in many real-world systems?

- A. Exploration can be costly or unsafe, rewards can be sparse or misaligned, and simulations may not match reality
- B. It does not need feedback
- C. It only works with spreadsheets
- D. It cannot learn sequences

Q29 - Very Hard - Model risk

A credit model is accurate overall but performs poorly for a small subgroup. Why is aggregate accuracy insufficient?

- A. It can hide subgroup harm and unfair performance differences
- B. It always implies fairness
- C. Small groups do not matter
- D. Accuracy is the same as recall

Q30 - Very Hard - Causal vs predictive

A model predicts that users who get discounts buy more. Why is this not enough to conclude discounts caused purchases?

- A. Discounts may have been targeted to users already likely to buy
- B. Classification models cannot use discounts
- C. Prediction always proves causality
- D. Data size eliminates confounding

Q31 - Very Hard - AI architecture

Which architecture is best for a regulated claims assistant that must cite policies and escalate uncertain cases?

- A. RAG over approved policy documents, confidence/grounding checks, audit logs, and human escalation
- B. A free-form chatbot with no retrieval
- C. A model trained only on random web text
- D. A UI without monitoring

Q32 - Very Hard - Data labeling

Why can inter-annotator agreement matter for supervised AI?

- A. Low agreement can indicate ambiguous labels, inconsistent guidelines, or unreliable ground truth
- B. It only affects file size
- C. It guarantees model fairness
- D. It removes the need for training

Q33 - Very Hard - AI economics

When should a simpler rules-based system be preferred over an ML system?

- A. When rules are stable, transparent, sufficient, cheaper, and lower-risk
- B. Never, ML is always better
- C. Only when there is unlimited data
- D. Only for image generation

Q34 - Very Hard - Generative AI governance

Why should organizations track AI-generated content provenance?

- A. To support auditability, accountability, IP review, and downstream trust
- B. To eliminate all storage costs
- C. To increase transparency
- D. To avoid all human review

Q35 - Very Hard - Operational safety

An AI assistant can trigger refunds. What control is most important?

- A. Permission checks, transaction limits, audit logs, confirmation steps, and human review for high-value cases
- B. Let the model decide without constraints
- C. Hide all actions from users
- D. Use the longest prompt possible

What is the purpose of model cards or similar documentation?

- A. Communicate intended use, limitations, evaluation results, risks, and ethical considerations
- B. Replace all tests
- C. Store API keys
- D. Train the model faster

Answer Key with Explanations

Q01. B - Systems that perform tasks associated with human intelligence such as reasoning, perception, learning, or language

Definitions / Medium

AI is a broad field covering systems that perform intelligence-like tasks.

Q02. A - ML is a subset of AI focused on learning patterns from data

AI vs ML / Medium

Machine learning is one approach within AI.

Q03. A - Use of multi-layer neural networks to learn representations

Deep learning / Medium

Deep learning uses layered neural networks to learn representations from data.

Q04. A - Explicit rules, logic, and knowledge representations

Symbolic AI / Medium

Symbolic AI represents knowledge through rules, symbols, and logic.

Q05. A - It creates new content such as text, images, code, or audio

Generative AI / Medium

Generative AI produces new artifacts rather than only predicting labels or scores.

Q06. A - A rule of thumb estimating distance or cost to a goal

Search / Medium

Heuristics guide search by estimating which paths are promising.

Q07. A - Perceives an environment and takes actions toward goals

Agents / Medium

Agents sense, decide, and act in an environment.

Q08. A - Sentiment analysis of reviews

NLP / Medium

NLP deals with natural language tasks such as sentiment analysis, translation, and summarization.

Q09. A - Detecting pedestrians in camera frames

Computer vision / Medium

Computer vision analyzes visual data such as images and video.

Q10. A - Feedback signal that indicates how good an action or outcome is

Reinforcement learning / Medium

Rewards guide an agent toward desirable behavior.

Q11. A - Models learn patterns from data, including errors and biases

Data quality / Medium

Poor data can produce poor, biased, or unreliable AI behavior.

Q12. A - Fairness, transparency, privacy, accountability, and safety

Responsible AI / Medium

Responsible AI considers societal, legal, and operational risks.

Q13. A - It represents entities and relationships explicitly for reasoning and retrieval

Knowledge representation / Hard

Knowledge graphs encode relationships that can support reasoning and structured retrieval.

Q14. A - Planning selects a sequence of actions to reach a goal; classification assigns labels

Planning / Hard

Planning is about action sequences, not just assigning a label.

Q15. A - It helps decide when to abstain, escalate, or request more information

Uncertainty / Hard

Uncertainty informs risk-aware behavior and human handoff.

Q16. A - Problem framing -> data -> modeling -> evaluation -> deployment -> monitoring

AI lifecycle / Hard

AI systems require problem definition, data, modeling, evaluation, deployment, and monitoring.

Q17. A - The metric may not align with the real decision, cost, or workflow

Evaluation / Hard

Model metrics must align with the operational objective and decision context.

Q18. A - High-stakes, uncertain, regulated, or irreversible decisions

Human-in-the-loop / Hard

Human review reduces risk where errors have serious consequences.

Q19. A - Humans over-trust automated recommendations even when they may be wrong

Automation risk / Hard

Automation bias is over-reliance on machine outputs.

Q20. A - Generated content can be plausible but unsupported or false

Generative AI risk / Hard

LLMs can generate fluent but incorrect content, so grounding and verification are needed.

Q21. A - AI behavior can degrade as data distributions change

AI operations / Hard

AI models depend on data distributions that can drift after deployment.

Q22. A - Users, auditors, or regulators may need to understand reasons behind decisions

Ethics / Hard

Explainability supports trust, debugging, accountability, and compliance.

Q23. A - Data minimization, access control, anonymization/pseudonymization where suitable, and retention limits

Data privacy / Hard

Privacy-preserving development limits unnecessary exposure of sensitive data.

Q24. A - They are large general-purpose models that can be adapted to many downstream tasks

Foundation models / Hard

Foundation models provide a base that can be prompted, fine-tuned, or adapted for many tasks.

Q25. A - It can combine learned perception/language with explicit rules, constraints, or reasoning

Hybrid AI / Very Hard

Hybrid designs can combine flexible learning with controllable rule-based constraints.

Q26. A - Bias testing, explainability, human review, documentation, privacy controls, and auditability

AI governance / Very Hard

High-impact employment use requires fairness, accountability, privacy, and audit controls.

Q27. A - Distribution shift between training and deployment conditions

Distribution shift / Very Hard

The deployment input distribution differs from training data, causing performance degradation.

Q28. A - Exploration can be costly or unsafe, rewards can be sparse or misaligned, and simulations may not match reality

RL tradeoffs / Very Hard

RL must balance exploration, safety, reward design, and sim-to-real transfer.

Q29. A - It can hide subgroup harm and unfair performance differences

Model risk / Very Hard

Aggregate metrics can mask poor subgroup performance.

Q30. A - Discounts may have been targeted to users already likely to buy

Causal vs predictive / Very Hard

Causal claims require controlling for confounding or using causal designs.

Q31. A - RAG over approved policy documents, confidence/grounding checks, audit logs, and human escalation

AI architecture / Very Hard

Regulated assistants need approved sources, grounding, traceability, and escalation.

Q32. A - Low agreement can indicate ambiguous labels, inconsistent guidelines, or unreliable ground truth

Data labeling / Very Hard

If humans disagree on labels, the model may learn noisy or inconsistent targets.

Q33. A - When rules are stable, transparent, sufficient, cheaper, and lower-risk

AI economics / Very Hard

The simplest system that meets requirements is often preferable.

Q34. A - To support auditability, accountability, IP review, and downstream trust

Generative AI governance / Very Hard

Provenance helps identify how content was produced and whether it can be trusted or reused.

Q35. A - Permission checks, transaction limits, audit logs, confirmation steps, and human review for high-value cases

Operational safety / Very Hard

Action-taking systems need strong controls, auditability, and approval for risky actions.

Q36. A - Communicate intended use, limitations, evaluation results, risks, and ethical considerations

Model documentation / Very Hard

Model documentation helps users understand appropriate use and limitations.

Last-minute revision checklist

- Differentiate AI, ML, deep learning, generative AI, symbolic AI, and foundation models.
- Understand search, planning, agents, knowledge representation, NLP, CV, and RL basics.
- Know AI lifecycle: framing, data, modeling, evaluation, deployment, monitoring.
- Explain responsible AI: fairness, privacy, transparency, accountability, and safety.
- Connect AI metrics to business value and workflow design.
- Recognize governance controls for high-impact systems.

Suggested timed drill

Pick 25 mixed questions and give yourself 15 minutes. Then review every wrong or guessed answer and write one sentence explaining the underlying concept.